List

A list is a collection of elements arranged in a sequential order. In memory, a list can be stored in two ways:

1. One way is we can store the Elements are stored in contiguous memory locations. This method is known as an array.
2. And the other way is we can use pointers or links to associate the elements sequentially. This method is known as a linked list.

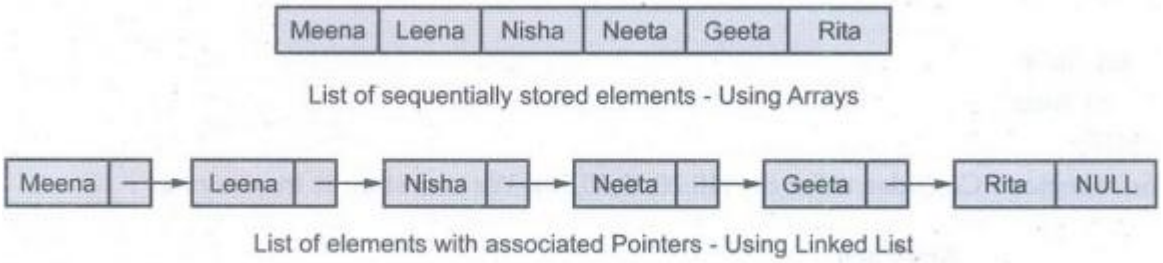

Fig. 1.3.1 Lists

Abstract Data Type (ADT) – List

In an ADT (Abstract Data Type), the implementation details are hidden. It only defines what operations can be performed, not how they are implemented.

AbstractDataType List

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Instances: A list is a collection of elements arranged in a linear (sequential) manner.

Operations: The various operations that can be performed on a list are:

1. **Insertion:** Used to insert an element into the list.
2. **Deletion:** Used to remove an element from the list.
3. **Searching:** Used to find an element based on a given key value.
4. **Modification:** Used to change the value of a specific element without changing its position.
5. **Display:** Used to display the elements of the list in forward or backward order.

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Implementation of List

A list can be implemented in two ways:

1. Array-based implementation of list

   Array of linked list is an important data structure used in many applications. It is an interesting structure to form a useful data structure. It combines static and dynamic structure. Static means air and dynamic linked list used to form a useful data structure this area of linked list structure is appropriate for application. Here, elements of list are stored in the (contiguous) cells of an array. The operation that are commonly performed in list are listed below:

   - Creating of a list
   - Inserting new element as a required position of list
   - Deletion of any element from list
   - Modification of any element of list
   - Traversing of list
   - Merging of two list

2. Linked list-based implementation of list